Franken FDK AAC — A Manual for the Audio Engineer

A complete guide to encoding audio in AAC / MPEG-4 and to all the options of this encoder. Written for someone who knows audio and is fluent with a computer, but who need not be a programmer or a mathematician.

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Part I. How AAC Actually Works

1. Why anyone needs a lossy codec

A PCM recording (what's on a CD or in a WAV file) stores every sample of sound exactly. At 44.1 kHz and 16 bits per channel in stereo, that's about 1411 kilobits per second. That's a lot. A lossy codec, such as AAC or MP3, can render the same material at 128, 96, or even 64 kilobits in such a way that the ear, under everyday conditions, hears no difference — or hears a very small one.

The trick is not about "squeezing" the sound like a ZIP file. It's about THROWING AWAY what you won't hear anyway, and storing only the rest. All the intelligence of a codec lies in guessing well what is irrelevant. That is the psychoacoustic model — a mathematical description of how human hearing works.

2. The three pillars of psychoacoustics

The encoder makes decisions based on a few phenomena worth understanding, because half the options in this program control exactly these.

SIMULTANEOUS MASKING. A loud sound at some frequency makes you stop hearing quieter sounds right next to it in the spectrum. If a loud 1000 Hz is playing, a quiet 1100 Hz signal is "masked" — the ear won't catch it. The encoder can therefore store that masked fragment very carelessly (few bits) or not at all. This is the most important bit-saving mechanism in AAC.

TEMPORAL MASKING. Just before and just after a loud hit (e.g. a snare), the ear is "deafened" and doesn't hear quiet details. The encoder exploits this.

THE THRESHOLD OF HEARING (ATH, Absolute Threshold of Hearing). There is a limit of silence below which the ear simply doesn't register sound — and this limit differs for different frequencies (we hear 2–5 kHz best, the edges of the band worse). Anything below that threshold can be thrown away.

When you hear that a codec "took away the air" or "flattened the cymbals," it almost always means that the psychoacoustic model deemed something inaudible, and your ear disagreed. A large part of this manual is learning how to shift those decisions toward your ear.

3. Bands, MDCT, and scale factors

AAC doesn't work on samples in time, but on the SPECTRUM. The signal is cut into frames (1024 samples) and transformed into the frequency domain (the MDCT transform). The resulting coefficients are grouped into SCALE FACTOR BANDS (scale factor bands, SFB for short). SFB numbering goes FROM THE BOTTOM: SFB 0 is the lowest frequencies (bass), and the higher the number, the higher up in the spectrum.

This matters, because many options in this encoder operate precisely on SFB band numbers. When we say "MS on the 5 highest bands," we mean the bands with the highest numbers.

In each band the encoder decides how precisely (how many bits) to store the coefficients. The fewer the bits, the larger the quantization error — audible as noise or a "hole" in the spectrum. The psychoacoustic model tells it where that error will hide under masking, and where it will be audible.

4. The AAC families: LC, HE-AAC, HE-AAC v2

AAC is not a single format, but a family of profiles. In this encoder you select them with the `-p` switch:

AAC-LC (Low Complexity, profile 2). The basic, most commonly used variant. It encodes the full band traditionally. Sensible from around 96 kbps stereo up. A safe choice for good-quality music.

HE-AAC (High Efficiency, profile 5, also called AAC+ or aacPlus). To LC it adds SBR — Spectral Band Replication. The AAC core encodes only the lower part of the band (e.g. up to 8–13 kHz), and the top is "recreated artificially" by SBR from the lower part and a small description. A huge bit saving while preserving the impression of a wide band. Sensible around 32–80 kbps stereo.

HE-AAC v2 (profile 29). To HE-AAC it adds PS — Parametric Stereo. Instead of encoding two channels, it encodes one (mono) plus a handful of parameters describing how to recreate stereo from it. Sensible at very low bitrates, 16–48 kbps stereo.

A practical rule: the lower the bitrate, the more "prostheses" (SBR, PS). At 320 kbps these prostheses only get in the way — use LC then.

5. What SBR is (the artificial upper band)

It's worth understanding SBR, because many options control it. SBR does NOT encode the upper band sample by sample. Instead it sends ENVELOPES — the information "in this piece of time and in this frequency range the energy has such and such a shape" — plus hints about which fragments of the lower band to "copy" upward, and how much noise to add. The decoder reconstructs the top from that.

That's why the upper band in HE-AAC sounds different from the original: it's an intelligent reconstruction, not a copy. The SBR options in this encoder let you control the density of the envelopes in time, their frequency resolution, the amount of injected noise, and so-called inverse filtering (more on that later).

6. What Parametric Stereo (PS) is

PS goes even further than SBR in saving. It sends one channel of sound and parameters describing the differences between left and right:

- IID (Inter-channel Intensity Difference) — the loudness difference between channels.
- ICC (Inter-channel Coherence) — how similar/coherent the channels are to each other.
- IPD/OPD (phase differences) — these are NOT supported in the FDK encoder (always zero); more on that honestly in the section on PS options.

From that one channel and those parameters the decoder "spreads" the sound back into stereo. At 24 kbps this is the only way for stereo to make any sense at all. At higher bitrates PS usually gets in the way and isn't used.

Part II. Bitrate, the Bit Reservoir, and Quality Control

7. CBR, VBR, and what really happens inside

CBR (Constant Bitrate) means a fixed bitrate: every second of sound takes up more or less the same amount of space. Convenient for streaming, predictable. You select it with the `-b` switch (e.g. `-b 128000` = 128 kbit/s).

VBR (Variable Bitrate) means a variable bitrate: difficult fragments (lots of detail, transients) get more bits, quiet and simple ones fewer. Theoretically better quality per bit. In FDK, VBR is selected with the `-m` switch (modes 1–5), but — and this must be said outright — VBR in FDK is weak. The presets are conservative and don't give the freedom that, say, LAME does in MP3.

The key thing to understand: EVEN in CBR mode the bitrate locally "breathes." The BIT RESERVOIR (bit reservoir) serves this. It's a buffer: when a frame is easy, the encoder stores it sparingly and puts the saved bits into the reservoir; when a hard frame comes, it borrows from the reservoir and spends more than would follow from the average. The average is preserved, but momentarily the bitrate jumps up and down. That's why "CBR" in AAC is not perfectly flat.

This encoder exposes reservoir control to the outside, and that is the basis of our own, better VBR — see the next chapter.

8. Recipe: a VBR that doesn't cut the last frames "by force"

This is exactly the situation you often mean: you have a difficult fragment (e.g. the entrance of a whole brass section or the cymbals), the encoder released a few frames well above the average — and instead of letting that quality "live," it starts choking the following frames just to get back to the average. The effect: an audible dimming right after the loud moment.

Our solution is a quasi-CVBR: we take the CBR engine (which keeps a sensible average), but we WIDEN the reservoir and the bit range so the encoder can breathe much harder and doesn't have to "repay the debt" immediately. Four switches:

`--vbr-reservoir <bits>` — the reservoir size. Larger = the encoder can hold elevated quality longer after a difficult fragment before returning to the average.

`--peak-bitrate <bps>` — the allowed momentary peak. It lets difficult frames reach high while preserving the average.

`--max-bits-frame <n>` and `--min-bits-frame <n>` — hard bit limits per frame. This is the core of "constrained VBR": you say "no fewer than X, no more than Y per frame."

`--bitres-mode <0|1|2>` — reservoir mode: 0 full, 1 reduced, 2 off (rigid bitrate frame by frame).

A PRACTICAL RECIPE for material that should "live" after difficult fragments (~128 kbps stereo, acoustic/orchestral music):

```
fdkaac-franken-x64.exe -p 2 -b 128000 --peak-bitrate 200000 --vbr-reservoir 12000 -o out.m4a in.wav
```

What it does: the average stays around 128, but when a climax comes, frames can reach 200 kbps, and the wide reservoir (12000 bits) means the encoder does NOT immediately choke the following frames — it lets the quality drop gently. This is exactly that "don't cut by force, let it live" effect.

Want it stronger: raise `--peak-bitrate` to 256000 and `--vbr-reservoir` to 18000. Want to hold the average more tightly (a link limit): reduce the reservoir to 4000.

AN IMPORTANT LIMITATION, honestly: this still works on the CBR engine + reservoir, not on true "by content" allocation like in LAME. The swings are moderate (on the order of ±20–50%), because AAC has a hard limit of 6144 bits per channel per frame. But this "light flight" is exactly what you usually want — and it is fully controlled.

9. Extreme bitrates: very low and very high

VERY LOW. By default FDK won't let you go below a certain floor. If you really want 8 kbps HE-AAC stereo or 6 kbps LC (even just for an experiment), use the `--unlock-bitrate` flag. WATCH OUT for one detail: normally this frontend treats small numbers with `-b` as kilobits (that is, `-b 96` = 96 kbps). In `--unlock-bitrate` mode it takes `-b` LITERALLY as bits per second — that is, `-b 8000` is 8000 bps = 8 kbps. Below roughly 10 kbps there's a natural floor resulting from the AAC headers themselves — you can't go lower without breaking the format.

```
fdkaac-franken-x64.exe -p 29 -b 8000 --unlock-bitrate -o out.m4a in48k.wav
```

TWO IMPORTANT CAVEATS for very low bitrates:

First — `--unlock-bitrate` lowers the floor only for AAC-LC (profile 2). HE-AAC and HE-AAC v2 (profiles 5 and 29) have THEIR OWN, hard floor of about 16 kbps, resulting from the fact that SBR cannot configure itself below that — the encoder will either raise the bitrate to 16 kbps or refuse to start at all. So if under HE-AAC you enter `-b 5` or `-b 8000`, you'll get 16 kbps (the program will warn you about this). Want to go truly lower than 16 kbps? Use AAC-LC: `-p 2 -b 8000 --unlock-bitrate`.

Second — WATCH the interpretation of `-b` in unlock mode. Normally this frontend treats small numbers as kilobits: `-b 128` = 128 kbps, `-b 1152` = 1152 kbps. But with `--unlock-bitrate` the number is taken LITERALLY as bits per second: `-b 8000` = 8000 bps = 8 kbps. If by mistake you wrote `-b 1152 --unlock-bitrate` (thinking "1152 kbps"), you'd get 1152 bps — practically nothing. The program will warn when a value looks like that mistake. Rule: in unlock mode always give the full number of bits (e.g. `-b 6000`, not `-b 6`).

VERY HIGH. The upper ceiling is 6144 bits per channel per frame. At 48 kHz that works out to about 576 kbps per channel; in stereo the real limit appears around 600 kbps total. If you want 1000+ kbps, you have to raise the sample rate above 48 kHz (a 96 kHz source) — then the ceiling rises and you'll reach 1024, even 1152 kbps. This is a limit of the AAC standard itself.

A note on unambiguity: at such high bitrates (1152) you do NOT use `--unlock-bitrate` (it serves for going down), so `-b 1152` there unambiguously means 1152 kbps. The interpretation collision applies only to unlock mode, where you're operating with small bps numbers anyway. In practice these two worlds don't touch.

9a. Is it worth doing an "afterburner plus" — more iterations at extreme bitrate

Since optimization at very high bitrate is your thing, one thing must be said outright, otherwise you'll waste time on a dead end. Inside the encoder there's a loop that iteratively picks the quantization (an internal parameter "maxIterations"; with the afterburner on it gives 5 tries, off 2). It's tempting to think "I'll give it 20 iterations, it'll be more precise." But this loop is a RESCUE mechanism for a bit SHORTAGE — it only kicks in when the frame doesn't fit in the budget and needs to be pressed down. At extremely HIGH bitrate the budget is enormous, the frame fits right away, so the loop almost never goes into further iterations. Raising the limit does nothing there — it's not "more polishing," but "a higher emergency-pressing limit," which you don't use anyway. The afterburner itself (`--afterburner 1`) already selects a more precise tuning table (and this is on by default) — and higher up this particular mechanism doesn't reach, because there are no additional levels in the codec.

What REALLY gives more detail at high bitrate is not the number of iterations, but LOWERING THE MASKING THRESHOLDS, so the encoder deems more things worth encoding at all: `--ath-scale` below 256, `--spread-mask` below 256, and `--ms-precision` above 256. These are the levers that actually convert surplus bits into preserved detail (see chapter 18). Quantization iterations don't.

VERY HIGH (continuation of the proper optimization) — see chapter 18.

Part III. Standard Options (Not Just Frankenstein)

These are the switches available in every version of this frontend. It's worth knowing them, because they're the foundation on which you build the rest.

10. Profile selection and basics

`-p <n>` — profile (Audio Object Type). The most important: `2` = AAC-LC, `5` = HE-AAC, `29` = HE-AAC v2, `23` = AAC-LD (low delay, for communication), `39` = AAC-ELD. For music: 2 for high bitrates, 5 for medium, 29 for the lowest.

`-b <n>` — bitrate in bits per second for CBR (e.g. `-b 192000`). Small numbers the frontend multiplies ×1000 (`-b 192` = 192 kbps) — outside `--unlock-bitrate` mode.

`-m <n>` — VBR mode 1–5 (1 lowest quality, 5 highest). As mentioned, weak in FDK; for serious work use CBR + chapter 8.

`-o <file>` — output file. The `.m4a` extension gives an MP4 container; `-f 2` switches to a raw ADTS stream (`.aac`).

`-w <n>` — the bandwidth of the core in Hz. Note: under SBR this does not work as you'd expect — to control the core band under HE-AAC use `--core-cutoff` (Part IV).

11. Afterburner and signaling

`--afterburner <0|1>` — the afterburner is an additional, slower quantization optimization algorithm. `1` = on (better quality, slower), and this is the default and recommended. Turn it off only when you care about speed.

`-s <n>` / signaling — how SBR/PS is signaled in the stream (compatibility with older decoders). The default "auto" is good in most cases.

12. Container and metadata

`--moov-before-mdat` — places the MP4 index at the start of the file (useful for progressive streaming). Standard tags (title, artist, etc.) are set with the `--title`, `--artist`, `--album` and related switches.

Part IV. Frankenstein Options — Controlling the Encoder's Insides

This is the heart of this encoder: switches that in ordinary FDK are hard-wired fixed, and here exposed to the outside. Each is described practically — what it does for the ear, when to use it, and what values make sense.

13. Stereo in the core: MS and Intensity

MS STEREO (Mid/Side). Instead of encoding the left and right channels separately, it encodes the sum (mid = L+R) and the difference (side = L−R). When the channels are similar (and in music they usually are), the difference is small and cheap to store. It's almost free saving.

`--msmask <0|1>` — force: `0` = separate L/R everywhere (full channel independence), `1` = MS on all bands. By default the encoder decides for itself per band.

`--msbands <n>` — restrict MS to bands with a number smaller than n. That is, it takes the N LOWEST bands (remember: bands are numbered from the bottom, 0 = bass). Above that — pure L/R. Example: `--msbands 6` = MS only on the 6 lowest bands.

`--msbands-lo <lo>` and `--msbands-hi <hi>` — this is a "FROM–TO" pair. You give BOTH switches and define the range of band numbers in which MS is allowed; outside that range it goes pure L/R. This lets you place MS ANYWHERE, including at the very top of the spectrum — which `--msbands` (always from the bottom) can't do.

How to remember it: `--msbands` = "from the bottom up to number N"; `--msbands-lo/-hi` = "from number LO to number HI." Examples (assuming about 49 bands in LC stereo):

- MS only on the 5 HIGHEST bands (merge the noise up top, leave the bottom in full stereo): the highest bands are numbers 44–48, so `--msbands-lo 44 --msbands-hi 48`.
- MS only in the middle of the band: `--msbands-lo 10 --msbands-hi 30`.
- MS on the 6 lowest: simpler `--msbands 6` (the same as `--msbands-lo 0 --msbands-hi 5`).

How many bands you really have for a given mode and frequency is shown by `--verbose` (the "active SFBs" field) — check the top number there before you set `--msbands-hi`.

`--ms-precision <n>` — the precision of bands encoded in MS, on the Q8 scale (256 = no change). Values above 256 lower the masking threshold on MS bands, so the encoder allocates MORE bits to their precise description — the "holes" in the MS spectrum become shallower. This is the equivalent of turning up quality in the LAME `-q` style. 384 is about 1.5×, 512 about 2×. There's no upper limit — it's useful at 512 kbps too. The most useful range is 48–192 kbps.

INTENSITY STEREO (IS). A more aggressive technique: for high bands, where the ear poorly localizes direction, the encoder sends one energy envelope instead of two channels. It saves a lot, but can narrow the scene.

`--is <0|1>` — globally turn IS on/off.

`--is-aggression <0..100>` — a CONSUMER slider for IS aggressiveness. This is what you usually want to use. 0 = FDK's default behavior (very cautious), 100 = maximally aggressive intensity. Why it's needed: FDK has three independent "gates" letting IS in (the bitrate-to-band ratio, the channel-correlation threshold, the minimum number of contiguous bands) and they must be met SIMULTANEOUSLY. Moving one threshold does nothing, because another blocks it. This slider moves all of them at once. Start at 40–70 and listen.

Advanced IS thresholds (for those experimenting — they act AFTER passing through the aggressiveness slider): `--is-min-sfbs <n>` (the minimum number of contiguous bands), `--is-corr-thresh <n>` (the channel-correlation threshold, Q8), `--is-lr-ratio <n>` (the L/R ratio threshold, Q8). A practical note: these thresholds act COUNTERINTUITIVELY — a lower correlation-threshold value means MORE aggressive IS, not the other way around.

14. Band and cutoff (including audiophile full band)

`--core-cutoff <Hz>` — the upper limit of the band encoded by the core, in Hz. It works also under SBR (unlike `-w`). E.g. `--core-cutoff 7500` under HE-AAC v2 gives the core 7.5 kHz, and SBR does the rest.

`--uncap-bandwidth` — audiophile. FDK has a hard-wired band limit for the core: the minimum of 20 kHz and half the sample rate. This means that EVEN with a 96 kHz source and a high bitrate, in reality nothing above 20 kHz was encoded. This flag removes the limit — `--core-cutoff` can then reach all the way up to half the sample rate. For 96 kHz material and listeners who want the full band:

```
fdkaac-franken-x64.exe -p 2 -b 320000 --uncap-bandwidth --core-cutoff 40000 -o out.m4a in96k.wav
```

Measured: without this flag there is practically zero energy above 20 kHz; with it the band up to 40 kHz is really encoded.

15. SBR — controlling the upper band

Overrides the settings from the internal SBR tuning table.

`--sbr-start <n>` / `--sbr-stop <n>` — the start and stop indices of the SBR band. NOTE: FDK validates the combination — a bad pair will give "encoder initialization failed." Pick proven pairs (e.g. for 64k stereo start=5 stop=9 works).

`--sbr-freqscale <n>` — frequency grouping (0 linear, higher = finer logarithmic). `--sbr-noise-bands <n>` — the density of the SBR noise description. `--sbr-amp-res <0|1>` — the amplitude resolution of the envelope (0 = 1.5 dB, 1 = 3 dB).

`--sbr-num-env <1|2|4>` — the number of envelopes per frame = the SBR time resolution. More envelopes = better-rendered changes in time in the upper band, at the cost of bits. IMPORTANT: this option FORCES a static time grid (it disables the transient detector), so on material with sharp attacks it may be worse. Good for stable, ringing sounds. (The value 8 exceeds the standard grid and is rejected.)

`--sbr-freqres-fixfix <0|1>` — the frequency resolution of the envelope.

`--sbr-stereo-mode <0..3>` — the stereo mode in the SBR layer (separate from core MS!): 0 = mono, 1 = LR (full, independent separation of left and right in the upper band), 2 = coupling (economical: shared envelope + level/balance), 3 = switch (by default the encoder chooses per frame). Force 1 for maximum stereo separation up top (audiophile), 2 for bit saving at low bitrate.

`--sbr-invf <0..3>` — inverse filtering (inverse filtering). It's a mechanism controlled by a TONALITY ESTIMATOR: SBR analyzes whether the upper band should be more tonal or more noise-like, and accordingly "whitens" the copied material. 0 = off, 1 = weak, 2 = medium, 3 = strong. Stronger = less "metallic" ringing up top, at the cost of some detail. Usually the automation controls it; force manually when you hear metallicness.

`--sbr-noise-floor-offset <n>` — the offset of the noise level injected by SBR. Larger values = more fill noise in the reconstruction.

16. Parametric Stereo (HE-AAC v2)

`--ps <0|1>` — force the sending of the IID parameter (loudness difference): 0 = never (flattens the stereo image to mono-like), 1 = always.

`--ps-iid-quant <0|1>` — the precision of IID quantization: 0 coarse, 1 fine.

`--ps-icc <0|1>` — force ICC (Inter-channel Coherence — channel coherence/similarity) on/off. `--ps-icc-mode <0|1>` — the ICC rotation mode (ROT_A / ROT_B).

HONESTLY about IPD/OPD (phase differences): the FDK encoder does NOT compute them. In the code it's explicitly marked as "not supported" — the phase fields are always zeroed. Exposing them would require writing inter-channel phase analysis from scratch, which is a large, risky task. That's why these parameters aren't here and can't be turned on.

17. TNS, PNS, and the afterburner

--tns-mask <n> / --tns-order <n> — TNS (Temporal Noise Shaping) shapes the quantization noise in time, so it hides under transients (important for sharp attacks, e.g. percussion). The mask is a bitmap of active filters, the order is the length of the filter.

--pns <0|1> — PNS (Perceptual Noise Substitution) replaces noise bands with the description "there's noise of such energy here" instead of encoding it precisely. A big saving on noise material.

--pns-start <Hz> — from what frequency PNS may act.

--force-pns — bypass the low-bitrate gate for PNS. EXPLANATION: FDK has a table that at very low bitrates (below about 28 kbps) completely disables PNS — and then --pns-start is ignored, because PNS doesn't work at all. That's why at 24 kbps you heard artifacts like from MP3, and at 64 kbps PNS worked. This flag bypasses the table and turns on PNS despite the low bitrate.

--ath-scale <n> — scaling of the threshold of hearing (ATH), in Q8 (256 = ×1.0). This is the GLOBAL masking regulator. Above 256 = raises the thresholds = the encoder deems more things inaudible = more aggressive, fewer bits. Below 256 = lowers the thresholds = the encoder preserves more detail = more bits. This is the simplest way to tell the encoder "be cleaner" or "be more economical."

Part V. Details at High Bitrate and Tuning for Speech

18. "More bits = more detail": how to really squeeze it out

A question that often comes up: since I'm giving 400 kbps, will the encoder really use it on a better description of details, or does part get wasted? In AAC there aren't such advanced, separate regulators as in MusePack (signal-to-mask ratio, tone-masks-noise, noise-masks-tone as separate knobs). But the effect "describe more precisely, mask less" we achieve with two tools that together do exactly the same thing.

--ath-scale <n> BELOW 256 — lowers the global threshold of hearing. The encoder stops assuming something is inaudible and starts encoding it precisely. This is the closest equivalent of lowering the masking threshold from MusePack. At 320–400 kbps try 200 or 180.

`--spread-mask <n>` BELOW 256 — controls the SPREADING of masking between neighboring bands. In psychoacoustics a loud band "spreads" its masking ability onto its neighbors (that's exactly tone-masks-noise). By reducing this parameter, you tell the encoder: assume less that neighboring bands mask each other — so more bands are treated as audible and precisely encoded. The biggest effect where bits are the limit (96–192 kbps); at very high bitrate the encoder has a surplus of bits anyway, so the change tends to be small.

An audiophile RECIPE (maximum detail, high bitrate, material with a rich top):

```
fdkaac-franken-x64.exe -p 2 -b 400000 --ath-scale 190 --spread-mask 128 -o out.m4a in.wav
```

The combination: a lowered global threshold (ath-scale 190) + reduced spreading of masking (spread-mask 128). The encoder preserves considerably more microdetail, which in a phase comparison with the original gives smaller differences. This is in the spirit of what MusePack did at 1000 kbps — within the bounds of what AAC allows exposing without rewriting the whole model.

For MS-ed material add `--ms-precision 448` — shallower holes in the MS bands are another place where high bitrate really gives better sound.

19. Tuning for human speech

Yes, in HE-AAC (specifically in the SBR layer) there's a separate tuning mode for speech. In ordinary FDK it's hard-wired as off — here we've exposed it with a flag:

`--speech` — turns on SBR tuning for human speech. It changes the inverse-filtering thresholds, the noise level, and disables parametric encoding of the upper band — all so that speech (podcast, audiobook, dialogue) at low bitrate sounds more natural, without "gurgling" up top. It applies to HE-AAC/HE-AAC v2 (because it works in SBR).

```
fdkaac-franken-x64.exe -p 5 -b 32000 --speech -o mowa.m4a podcast.wav
```

Honestly about LC: pure AAC-LC (without SBR) does NOT have a separate, distinct "speech mode" in FDK. The LC psychoacoustic model is the same for speech and music. For speech in LC it's best to simply lower the band (`--core-cutoff` e.g. 12000–14000 Hz, because speech doesn't have much above that) and possibly turn on `--pns` with `--force-pns` at very low bitrates. But there's no dedicated "speech" switch for LC, because it's not in the codec itself — and I don't want to pretend it is.

Part VI. Ready-Made Recipes (From Extreme to Extreme)

20. A set of practical recipes

MUSIC, HIGH ARCHIVAL QUALITY (transparent stereo):

```
fdkaac-franken-x64.exe -p 2 -b 256000 --afterburner 1 -o out.m4a in.wav
```

MUSIC, "BREATHING" QUASI-VBR (doesn't cut after climaxes):

```
fdkaac-franken-x64.exe -p 2 -b 160000 --peak-bitrate 256000 --vbr-reservoir 16000 -o out.m4a in.wav
```

AUDIOPHILE, FULL BAND 96 kHz + maximum detail:

```
fdkaac-franken-x64.exe -p 2 -b 400000 --uncap-bandwidth --core-cutoff 40000 --ath-scale 190 --spread-mask 128 -o out.m4a in96k.wav
```

MUSIC STREAMING, MEDIUM BITRATE:

```
fdkaac-franken-x64.exe -p 5 -b 64000 --sbr-stereo-mode 3 -o out.m4a in.wav
```

PODCAST / SPEECH, LOW BITRATE:

```
fdkaac-franken-x64.exe -p 5 -b 32000 --speech --core-cutoff 12000 -o out.m4a mowa.wav
```

EXTREMELY LOW (experiment, 8 kbps stereo):

```
fdkaac-franken-x64.exe -p 29 -b 8000 --unlock-bitrate -o out.m4a in48k.wav
```

RESCUING STEREO AT LOW BITRATE (aggressive intensity):

```
fdkaac-franken-x64.exe -p 2 -b 96000 --is 1 --is-aggression 70 -o out.m4a in.wav
```

MS ONLY UP TOP (bottom full stereo, top merged):

```
fdkaac-franken-x64.exe -p 2 -b 128000 --msbands-lo 44 --msbands-hi 48 -o out.m4a in.wav
```

21. How to approach your own experiments

Change ONE parameter at a time and listen (or compare the spectrum/phase with the original). The encoder with no Frankenstein flags behaves exactly like the original FDK — that's your reference point. Every flag is a deliberate deviation from the default, proven behavior.

The order worth trying when there are problems:

- "too little detail / too flat" → `--ath-scale` down, then `--spread-mask` down.
- "stereo too narrow up top" → `--sbr-stereo-mode 1` (HE-AAC) or less aggressive IS.
- "metallic highs" → `--sbr-invf 2` or 3.
- "cuts after loud fragments" → `--peak-bitrate` + `--vbr-reservoir` (chapter 8).
- "artifacts like MP3 at very low bitrate" → `--force-pns`.
- "I want the full band from 96 kHz" → `--uncap-bandwidth --core-cutoff 40000`.

22. A closing note on integrity

All the behaviors described here were verified by measurement (stream decodability, real bitrate spread, spectral content), not just assumed. Where something has a limitation (IPD/OPD unsupported, no speech mode in LC, the ceiling of 6144 bits per channel, a residual bitrate floor of ~10 kbps), it is said outright, instead of promising things the codec can't do.

The encoder without flags = pure FDK. The flags = your deliberate decisions. Happy tuning.

Part VII. Reference Tables

Three orientation tables computed from the tuning tables in the FDK code. They are APPROXIMATE (the band grid is stepped), but they show the right order of magnitude and help you deliberately choose settings without guessing.

23. Table 1 — the upper frequency of the band (SFB)

The spectrum is divided into bands (scale factor bands, SFB), numbered from the bottom: band 0 is the lowest bass, the higher the number, the higher in frequency. When you restrict something to "N bands" (`--msbands`, `--isbands`), this table tells you what frequency that roughly corresponds to. Every fourth band is shown; the last row is the total number of bands and the Nyquist frequency (half the sample rate).

SFB	16 kHz	22.05 kHz	32 kHz	44.1 kHz	48 kHz	96 kHz
0	62	43	62	86	94	188

SFB	16 kHz	22.05 kHz	32 kHz	44.1 kHz	48 kHz	96 kHz
4	312	215	312	431	469	938
8	562	388	562	775	844	1688
12	875	646	1000	1378	1500	2438
16	1250	991	1500	2067	2250	3750
20	1656	1335	2250	3101	3375	5625
24	2188	1852	3375	4651	5062	8062
28	2875	2584	5000	6891	7500	12938
32	3844	3618	7000	9647	10500	24000
36	5188	5039	9000	12403	13500	36000
40	7000	7020	11000	15159	16500	48000
44	—	9647	13000	17916	19500	—
48	—	—	15000	22050	24000	—
number of bands / Nyquist	43 / 8000	47 / 11025	51 / 16000	49 / 22050	49 / 24000	41 / 48000

An example of how to read it: at 44.1 kHz band no. 40 ends around 15.2 kHz. If you want MS to work only below ~15 kHz, set `--msbands 40`.

24. Table 2 — SBR: the start index vs. the crossover frequency

In HE-AAC the lower part of the band is encoded by the AAC core, and the upper part is added by SBR. `--sbr-start` (index 0..15) decides at what frequency SBR begins — a lower index means SBR comes in lower, so the core is narrower. "core" is the core's sample rate; in dual-rate mode the output is twice as high as the core.

start index	core 16 kHz	core 24 kHz	core 32 kHz	core 44.1 kHz	core 48 kHz
0	2750	2250	2500	1378	1500
1	3000	2625	3000	2067	2250
2	3250	3000	3500	2756	3000
3	3500	3375	4000	3445	3750
4	3750	3750	4500	4134	4500
5	4000	4125	5000	4823	5250
6	4250	4500	5500	5512	6000

start index	core 16 kHz	core 24 kHz	core 32 kHz	core 44.1 kHz	core 48 kHz
7	4500	4875	6000	6202	6750
8	4750	5250	6500	6891	7500

The upper limit of SBR is set by `--sbr-stop` (0..13) — a higher index means SBR reaching higher. By default the library picks both indices for the bitrate; give these values only when you deliberately want to shift the crossover point.

25. Table 3 — AAC-LC: automatic band cutoff by bitrate

When you don't give `-w`, the encoder itself picks the bandwidth from this table — by bitrate PER CHANNEL (128 kbps stereo is 64 kbps per channel). The table is common for 32/44.1/48 kHz and higher; the sample rate affects only the upper limit (Nyquist).

bitrate per channel	mono band	stereo and more band
up to 12 kbps	3700	5000
20 kbps	6900	9640
28 kbps	9600	13050
40 kbps	12060	14260
56 kbps	13950	15500
72 kbps	14200	16120
96 kbps and more	17000	17000

A practical conclusion: in pure AAC-LC (without SBR) the automation stops around 17 kHz anyway. Giving `-w` higher makes sense only with a large bitrate reserve; a full band above 20 kHz requires `--uncap-bandwidth` and a sample rate of at least 96 kHz.

26. How to read `--verbose`

`--verbose` prints raw values without hints in parentheses, so the readout is clean. Here's what the less obvious ones mean:

- AOT (profile): 2 = AAC-LC, 5 = HE-AAC, 29 = HE-AAC v2, 23 = AAC-LD, 39 = AAC-ELD.
- bitrate-mode: 0 = CBR, 1 to 5 = VBR (higher = better quality).
- channel-mode: 1 = mono, 2 = stereo. In HE-AAC v2 the core is mono, and stereo is recreated from parameters (Parametric Stereo).
- core bandwidth: the upper frequency of the AAC core. The SOURCE is given in parentheses: "from -w" (you gave it manually), "from --core-

cutoff" or "auto". Under SBR this is only the core — SBR plays above this value.
- final BW (AAC+SBR): shown only with SBR active — the approximate upper frequency of the WHOLE signal (core plus SBR), computed from the SBR stop index. It's a complement to "core bandwidth": that says how far the core alone reaches, and this how far the whole HE-AAC reaches after adding SBR.
- sbr-ratio: 1 = single-rate (downsampled), 2 = dual-rate (the core at half the output frequency).
- sbr amp res: 0 = resolution 1.5 dB, 1 = 3.0 dB.
- codec delay: the codec delay in samples per channel — important for gapless.
- IS corr threshold, IS L/R ratio: thresholds on the Q8 scale, where 256 = 1.0. A lower correlation threshold means intensity stereo turns on MORE READILY (contrary to intuition).
- franken overrides applied: a list of the switches that in this run deviate from pure FDK (or "none," when you changed nothing).

Values on the Q8 scale (like the IS thresholds, `--ms-precision`, `--ms-bias`, `--ath-scale`, `--spread-mask`) are numbers where 256 means 1.0 — for example 243 is about 0.95.

Part VIII. Glossary

For quick recall — all the terms from this manual, explained in one or two sentences, in the language of the audio engineer.

AAC (Advanced Audio Coding). A family of lossy audio codecs from the MPEG family, the successor to MP3. Better quality per bit than MP3, especially at low bitrates.

AAC-LC (Low Complexity). The basic AAC profile. Encodes the full band classically. The choice for high bitrates and archiving.

Afterburner. An additional, slower algorithm optimizing quantization in FDK. On, it gives slightly better quality. On by default and recommended.

ATH (Absolute Threshold of Hearing). The threshold of hearing — the limit of silence dependent on frequency, below which the ear doesn't register sound. The encoder throws away what is below it. In this encoder regulated by `--ath-scale`.

Bitrate. The amount of data per second of sound, in kilobits (kbps) or bits (bps). Higher = more room for details = better quality (up to a certain limit).

Bit reservoir. A buffer allowing bits to be borrowed between frames. Thanks to it even CBR locally "breathes." The basis of our quasi-VBR.

CBR (Constant Bitrate). A fixed bitrate. Predictable size, convenient for streaming. In AAC slightly variable anyway thanks to the reservoir.

Coupling (SBR). A stereo mode in the SBR layer in which both channels share a common envelope plus a level description — economical, but a narrower scene in the upper band.

Cutoff. The upper limit of the band encoded by the core. Above it either silence, or (in HE-AAC) SBR takes over the job.

HE-AAC (High Efficiency). AAC-LC + SBR. The core encodes the bottom of the band, SBR recreates the top. For medium and low bitrates.

HE-AAC v2. HE-AAC + Parametric Stereo. For the lowest stereo bitrates.

ICC (Inter-channel Coherence). A PS parameter describing how similar/coherent the channels are to each other. Controlled by `--ps-icc`.

IID (Inter-channel Intensity Difference). A PS parameter describing the loudness difference between channels. The primary carrier of the stereo image in PS.

Intensity Stereo (IS). A technique in which high bands are encoded as one energy envelope instead of two channels. Economical, but can narrow the scene. Controlled by the `--is-aggression` slider.

Inverse filtering. An SBR mechanism "whitening" the copied material of the upper band, controlled by a tonality estimator. Stronger = less metallicness. Regulated by `--sbr-invf`.

IPD/OPD (phase differences in PS). NOT supported in the FDK encoder (always zero).

Quantization. Rounding spectrum coefficients to a finite precision. The fewer the bits, the larger the error (noise/holes). The heart of lossy compression.

Masking. The phenomenon in which a loud sound makes you not hear quieter ones right next to it (in the spectrum or in time). The main source of bit saving in AAC.

MDCT. A transform carrying a frame of sound from time to the frequency domain. The whole psychoacoustic model works on its result.

MS Stereo (Mid/Side). Encoding the sum (mid) and difference (side) instead of L and R. Almost free saving when the channels are similar.

PNS (Perceptual Noise Substitution). Replaces noise bands with the description "there's noise of such energy here" instead of encoding them precisely. Controlled by `--pns`, `--force-pns`.

Parametric Stereo (PS). Encodes one channel + parameters (IID, ICC), from which the decoder recreates stereo. For the lowest bitrates.

Profile (AOT, Audio Object Type). The AAC variant selected by `-p` (LC, HE, v2, LD, ELD).

Frame. A portion of sound (1024 samples) that the encoder processes at once.

SBR (Spectral Band Replication). Reconstruction of the upper band from the lower plus a small description. The heart of HE-AAC.

SFB (Scale Factor Band). A scale factor band. A group of neighboring spectrum coefficients for which the encoder makes a common decision about precision. Numbered from the bottom (0 = bass).

Spreading (of masking). The propagation of masking ability onto neighboring bands. Reduced by `--spread-mask` for more detail.

Tuning table. A set of default SBR/PS settings chosen by FDK by bitrate and frequency. Our flags override it.

TNS (Temporal Noise Shaping). Shapes the quantization noise in time, so it hides under transients. Important for percussion and sharp attacks.

Transient. A sharp, short sound (a hit, an attack). Difficult for the codec, because the quantization noise can "outrun" the attack (pre-echo).

VBR (Variable Bitrate). A variable bitrate dependent on the difficulty of the material. In FDK weak; our quasi-VBR (chapter 8) is a better alternative.

End of glossary.